

----- Active space -----

NECROTIZING ENTEROCOLITIS AND NEONATAL SEPSIS

m/c and devastating surgical emergency in neonates.

Etiopathogenesis

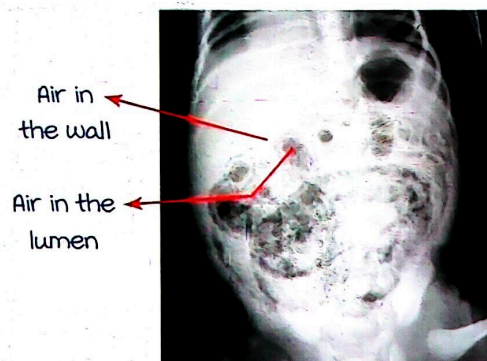
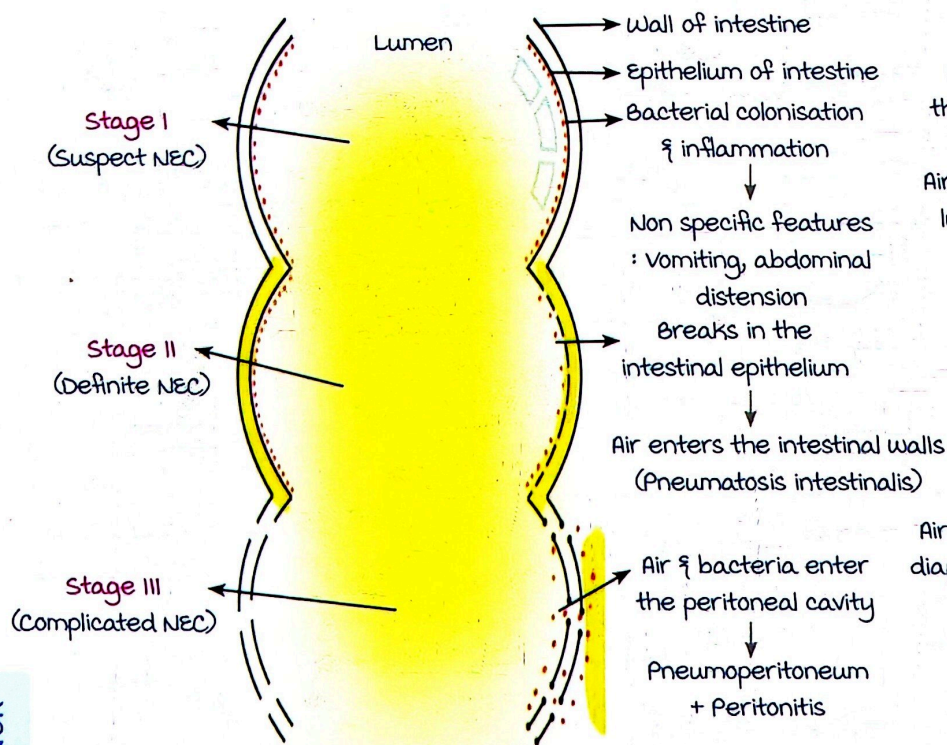
00:00:30

1. Prematurity (<32 weeks) → Immaturity of GI tract → **Inflammatory necrosis** d/t infection.
2. Top feeds without breastmilk → Pathogenic bacterial colonisation
(Eg: Formula, cow milk)
3. Other factors :
 - Asphyxia
 - maternal cocaine use
 } vasoconstriction of splanchnic circulation → **Necrosis.**

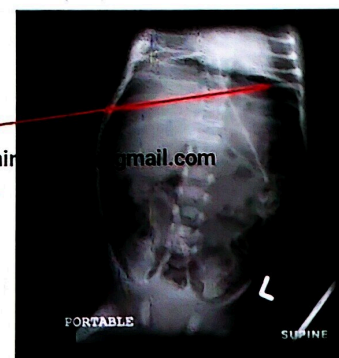
Clinical Features

00:03:00

MODIFIED BELL'S STAGING :



II a : Pneumatosis intestinalis



III b : Pneumoperitoneum

Paediatrics • v1.0 • Marrow 8.0 • 2024

Page 1/5

	Features	Imaging
I A	Occult blood in stools	Normal
I B	Gross blood in stools	
II a	Absent bowel sounds	Pneumatosis intestinalis (Characteristic of NEC)
III b	Absent bowel sounds + abdominal wall edema. khanirfan0392@gmail.com	Air in the intestinal wall ↓ Enters venous drainage system ↓ Superior mesenteric vein ↓ Portal vein : Pneumatosis portalis
III a	Induration Redness/erythema Tenderness } Of abdominal wall	-
III b	Intestinal perforation	Pneumoperitoneum

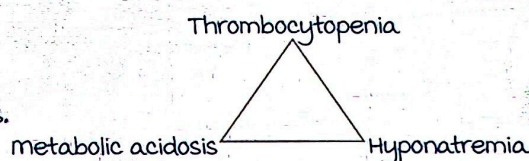
----- Active space -----

Management

00:15:28

Triad of lab findings :

- Seen only **after stage II b.**
- Do not have great relevance in diagnosis.

**until stage III a :****medical management :**

- Nil per oral (NPO).
- Total parenteral nutrition (TPN).
- IV antibiotics :
 - Penicillin (Gram +ve) + Aminoglycosides (Gram -ve) + metronidazole (Anaerobes).

Duration of antibiotics :

- Stage I a : 3 days.
- Stage I b : 7-10 days.
- Stage II, III a : 14 days.

Stage III b :Hemodynamically unstable : 1^o peritoneal drainage (Emergency procedure)↓
After stabilization

Hemodynamically stable :

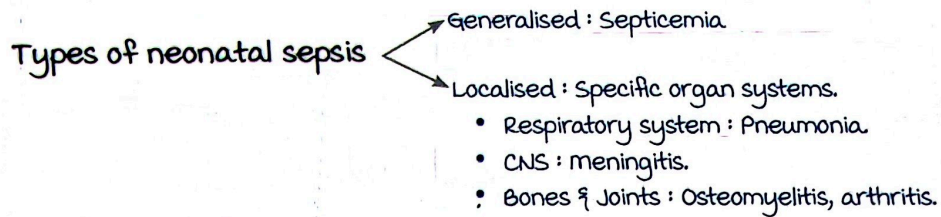
↓
Laparotomy → Resection & anastomosis.

Untreated : m/c cause of neonatal mortality.

----- Active space -----

Spectrum of neonatal sepsis

00:20:28



	Early onset sepsis	Late onset sepsis
Onset	First 72 hrs of life	> 72 hrs
Source of infection	maternal	Hospital acquired infection
Risk Factors	• Chorioamnionitis. • Infection of maternal genitalia (Foul smelling liquor). • Rupture of membranes over > 72 hrs.	• unclean hands of health professionals & caregivers. • Low birth weight (LBW) → ↓ Immunity. • Lack of breastfeeding.
Type of infection	Pneumonia >> Sepsis	Equal incidence of Sepsis, Pneumonia & meningitis

Etiology & Clinical Features

00:25:17

ORGANISMS IMPLICATED :**India :**

1. **Klebsiella** (m/c).
2. Acinetobacter.
3. S. aureus.

Worldwide :

1. Group B Streptococcus (GBS).
2. E. Coli.
3. Listeria.

SYMPTOMS :

Weak immune response → **Non specific symptoms** → Suspect sepsis in any sick baby.

Recognised symptoms :**Early symptoms :**

- Change in feeding pattern.
- Lethargy.
- Hypothermia >> Fever.

Pneumonia :

- Respiratory distress.

Signs specific to organ system :**meningitis :**

- Fever and irritability (Non specific).
- Seizures.

Management

648c85cfee3b03a74e182fab

00:29:23

----- Active space -----

INVESTIGATIONS :Blood culture : **Gold standard**.

Disadvantage : Time-consuming.

Sepsis screen : Presence of > 2 findings → The sepsis screen.

Components	Value in Sepsis
Total leukocyte count	↓ (< 5000/mm ³)
Absolute neutrophil count	↓ (< 1800/mm ³)
Immature/total neutrophil ratio	↑ (> 0.2)
micro-ESR	↑ (> 15 mm/hr)
CRP	↑ (> 1mg/dL)

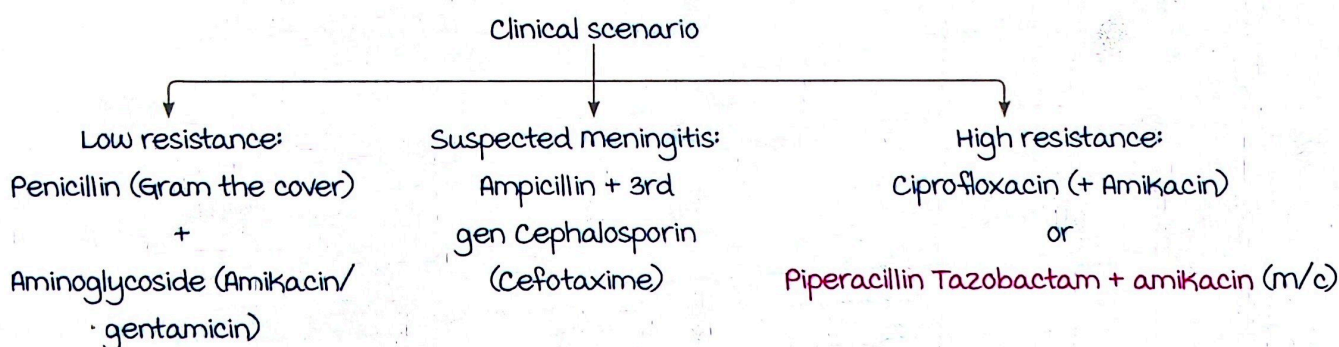
Newest marker for neonatal screening : ↑ **procalcitonin**.

Lumbar puncture :

- Indication : Suspicion of meningitis in all **symptomatic** babies.
- C/I : Asymptomatic babies (even with risk factors).

TREATMENT :

Empirical antibiotics (While awaiting culture reports):



Duration of treatment :

1. Culture negative : 5-7 days (1 week).

2. Culture positive :

- meningitis : 3 weeks
- Others : 2 weeks

Sepsis Algorithm

648c85cfee3b03a74e182fab

----- Active space -----

